Introduction to Quantum Computing

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1 Basics concepts

Classical computing is based on the concept of bits, which can take values of either 0 or 1. **Quantum bits**, or **qubits**, on the other hand, can exist in a superposition of 0 and 1.

A quantum state $|\psi\rangle$ is described as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1.$$

The ket notation of $|\psi\rangle$ stands for

$$|\psi\rangle \in \mathbb{C}^2, \quad |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \implies |\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

Difference from Classical Bit Unlike a classical bit, a qubit cannot be directly observed: the amplitudes α and β cannot be measured. A standard measurement will result in either 0 or 1, with probabilities $|\alpha|^2$ and $|\beta|^2$, respectively.

Additionally, the measurement in general **changes** the qubit state ("wavefunction collapse"). For example, if we measure $|\psi\rangle$ and get 0, the state collapses to $|0\rangle$, and if we get 1, it collapses to $|1\rangle$.

In practise we can estimate the probabilities $|\alpha|^2$ and $|\beta|^2$ by repeating the same experiment many times (outcome statistics). These repetitions are called **shots** or **trials**.

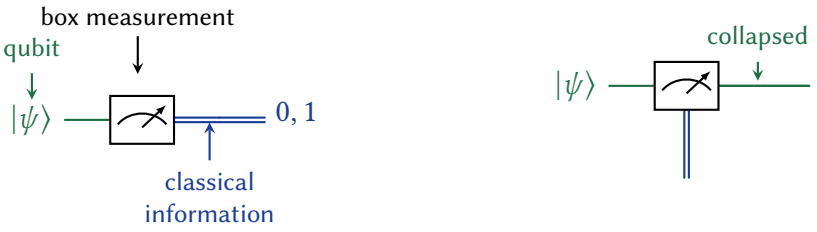


Figure 1: Circuit notation for a measurement.

What is a qubit physically? Many poossible realizations of qubits are>

- Two different polarizations of a photon (e.g.,left/right circular)
- Alignment of nuclear or electron spin ↑↓
- Ground state or excited state of an atom (electronic state)
- Clockwise or counterclockwise current in a superconducting loop with Josephson junctions (flux qubits)

Useful graphical representations of a qubit state are the **Bloch sphere**: if α and β happen to be real-valued that we can find angle $\theta \in \mathbb{R}$ such that $\alpha = \cos(\theta/2)$ and $\beta = \sin(\theta/2)$

$$|\alpha|^2 + |\beta|^2 = \cos^2(\theta/2) + \sin^2(\theta/2) = 1.$$

In general, α and β are complex numbers. Using so-called phase angles γ for α and $\gamma + \varphi$ for β, we represent

$$\alpha = e^{i\gamma}\cos(\theta/2), \qquad \beta = e^{i(\gamma+\varphi)}\sin(\theta/2), \quad \theta \in [0, \pi], \ \varphi \in [0, 2\pi).$$

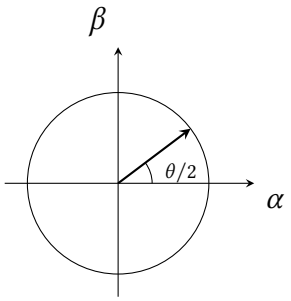


Figure 2: Real-valued α, β: the state is a point on the unit circle.

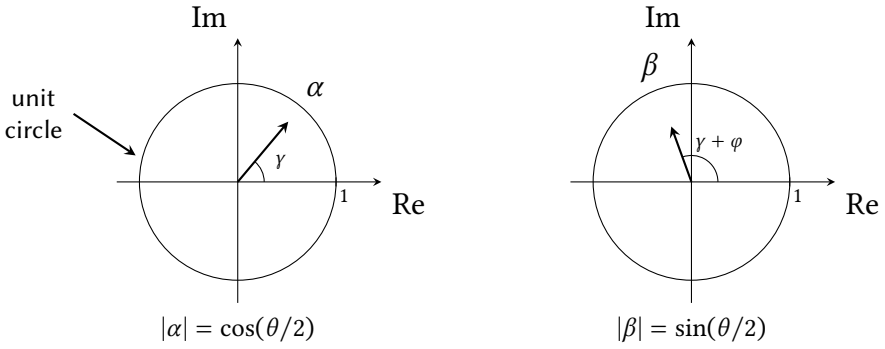


Figure 3: Complex planes of α and β.

Then

$$\begin{aligned} |\psi\rangle &= e^{i\gamma}\cos(\theta/2)|0\rangle + e^{i(\gamma+\varphi)}\sin(\theta/2)|1\rangle \\ &= e^{i\gamma}\left(\cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle\right) \end{aligned}$$

thus, up to the global phase $e^{i\gamma}$ (which is not observable), $|\psi\rangle$ is characterized by two angles θ and φ and these specify the point on the **Bloch sphere** defined as:

$$\vec{r} = \begin{pmatrix} \cos(\varphi)\sin(\theta) \\ \sin(\varphi)\sin(\theta) \\ \cos(\theta) \end{pmatrix} \in \mathbb{R}^3$$

On the surface of a sphere:

2 Single-qubit gates

2.1 Time evolution and circuit notation

Principle of time evolution The quantum state evolves over time. A quantum state $|\psi\rangle$ at a current time t transitions to a new quantum state $|\psi'\rangle$ at a later time point $t' > t$. The transition is described by a complex unitary matrix $\mathcal{U} \in \mathbb{C}^{2\times 2}$:

$$|\psi'\rangle = \mathcal{U}|\psi\rangle$$

a matrix vector multiplication.

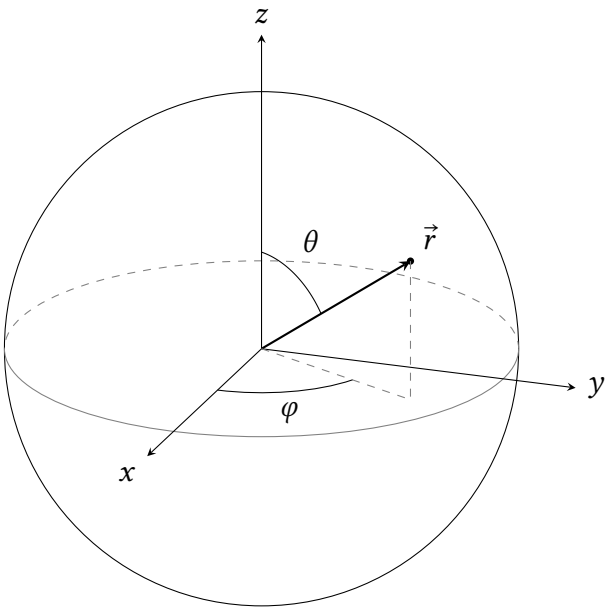


Figure 4: Bloch sphere: the point $\vec{r}$ is specified by the angles θ and φ.

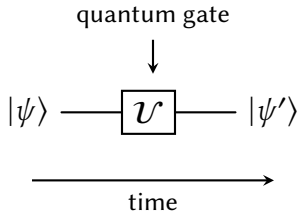


Figure 5: Circuit notation for a single-qubit gate.

Circuit Notation The circuit is read from left to right, but matrix times vector multiplication is from right to left ($\mathcal{U}|\psi\rangle$). Additionally, $\mathcal{U}$ preserves normalization: since $\mathcal{U}$ is unitary ($\mathcal{U}^\dagger\mathcal{U} = \mathbb{I}$), $\|\mathcal{U}|\psi\rangle\|^2 = \langle\psi \,|\, \mathcal{U}^\dagger\mathcal{U} \,|\, \psi\rangle = \langle\psi \,|\, \psi\rangle = 1$.

2.2 Common gates

Example 2.1 (A quantum analog of a classical NOT gate)**.** The classical NOT gate flips a bit $0 \mapsto 1$ and $1 \mapsto 0$. The quantum analog of a NOT gate is the **Pauli-X gate**, which flips the quantum state $|0\rangle \mapsto |1\rangle$ and $|1\rangle \mapsto |0\rangle$. The Pauli-X gate is represented by

$$X \equiv \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

$$X|0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle,$$
$$X|1\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$

In addition we define the **Pauli-Y gate**

$$Y \equiv \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},$$

and the **Pauli-Z gate**

$$Z \equiv \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Z leaves $|0\rangle$ unchanged, but flips the sign of the coefficient of $|1\rangle$. We recall the **Bloch sphere representation**

$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle,$$

then

$$\begin{aligned} Z|\psi\rangle &= \cos(\theta/2)|0\rangle - e^{i\varphi}\sin(\theta/2)|1\rangle \\ &= \cos(\theta/2)|0\rangle - \underbrace{e^{i\pi}e^{i\varphi}}_{e^{i(\pi+\varphi)}}\sin(\theta/2)|1\rangle \end{aligned}$$

remembering that $e^{i\pi} = -1$. New bloch sphere coordinates are $\theta' = \theta$ and $\varphi' = \pi + \varphi$, so the Z gate corresponds to a rotation of the Bloch sphere around the z-axis by π.

Proof that $e^{i\pi} = -1$. Recall the power series, valid for all $z \in \mathbb{C}$:

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \qquad \cos x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}, \qquad \sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}.$$

Let $x \in \mathbb{R}$ and set $z = ix$. Since $i^{2k} = (-1)^k$ and $i^{2k+1} = i\,(-1)^k$, splitting the series into even and odd terms (allowed by absolute convergence) gives

$$e^{ix} = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} + i \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} = \cos x + i \sin x.$$

Evaluating at $x = \pi$, where $\cos \pi = -1$ and $\sin \pi = 0$, we obtain

$$e^{i\pi} = \cos \pi + i \sin \pi = -1 + 0 = -1. \qquad \square$$

X, Y, Z gates are also called **Pauli matrices** and are Hermitian, i.e. $X^\dagger = X$, $Y^\dagger = Y$, $Z^\dagger = Z$. They are also unitary, i.e. $X^{-1} = X^\dagger$, $Y^{-1} = Y^\dagger$, $Z^{-1} = Z^\dagger$.

The **Pauli Vector** is defined as $\vec{\sigma} = (\sigma_1, \sigma_2, \sigma_3) = (X, Y, Z)$ is a vector of 2×2 matrices.

Hadamard Gate We define the **Hadamard gate** H as

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

so that

$$\begin{aligned} H|\psi\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{\sqrt{2}}(\alpha + \beta) \\ \frac{1}{\sqrt{2}}(\alpha - \beta) \end{pmatrix} \\ &= \frac{1}{\sqrt{2}}(\alpha + \beta)|0\rangle + \frac{1}{\sqrt{2}}(\alpha - \beta)|1\rangle \\ &= \alpha \frac{|0\rangle + |1\rangle}{\sqrt{2}} + \beta \frac{|0\rangle - |1\rangle}{\sqrt{2}} \end{aligned}$$

$$\alpha|0\rangle + \beta|1\rangle \quad \boxed{H} \quad \alpha\,\frac{|0\rangle + |1\rangle}{\sqrt{2}} + \beta\,\frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Figure 6: Hadamard gate on a general state.

Phase Gate: S and T We define the **phase gate** S as

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \implies S^2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z$$

and the **phase gate** T as

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} \implies T^2 = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{pmatrix} = S$$

since $(e^{i\pi/4})^2 = e^{i\pi/2} = i$.

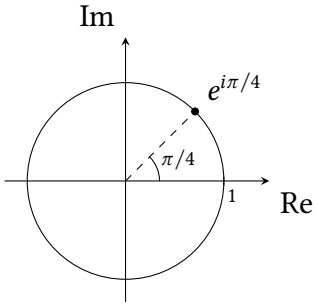


Figure 7: Phase of the T gate.

Remark. The Pauli matrices satisfy the following relations. Since each σ_j is Hermitian and unitary,

$$\sigma_j^2 = \sigma_j \sigma_j^\dagger = \mathbb{I} \quad \text{for } j = 1, 2, 3, \quad \text{i.e. } X^2 = Y^2 = Z^2 = \mathbb{I}.$$

Additionally, the Pauli matrices anticommute:

$$\sigma_j \sigma_k = -\sigma_k \sigma_j, \quad \forall j \neq k$$

We define the commutator of two matrices A and B as

$$[A, B] := AB - BA$$

and for the Pauli matrices we have

$$[\sigma_j, \sigma_k] = \sigma_j \sigma_k - \sigma_k \sigma_j = 2i\sigma_l$$

for (j, k, l) a cyclic permutation of $(1, 2, 3)$ e.g. $j = 2, k = 3, l = 1$. So for example,

$$[\sigma_2, \sigma_3] = 2i\sigma_1$$
$$[Y, Z] = 2iX$$

where the cyclic permutations of $(1, 2, 3)$ are $(1, 2, 3)$, $(2, 3, 1)$, $(3, 1, 2)$.

2.3 The matrix exponential

General definition of the **matrix exponential** (via power series expansion):

$$e^A \equiv \exp(A) = \sum_{k=0}^{\infty} \frac{1}{k!} A^k, \quad A \in \mathbb{C}^{n\times n}$$

with $A^k = A \cdot A \cdots A$ (k times) and $n \in \mathbb{N}$.

Generalization of Euler's formula Special case: self-inverse matrix $A^2 = \mathbb{I}$ (identity), and $x \in \mathbb{R}$ a real number. Then

$$e^{iAx} = \cos(x)\mathbb{I} + i\sin(x)A,$$

which generalizes Euler's formula $e^{ix} = \cos(x) + i\sin(x)$.

Proof. Since $A^2 = \mathbb{I}$, it is clear that $A^k = \begin{cases} \mathbb{I} & k \text{ even} \\ A & k \text{ odd} \end{cases}$. Therefore, the power series expansion of the exponential can be partitioned as

$$\begin{aligned} e^{iAx} &= \mathbb{I} + iAx - \frac{1}{2!}\mathbb{I}x^2 - \frac{i}{3!}Ax^3 + \ldots \\ &= \mathbb{I}\left(1 - \frac{1}{2!}x^2 + \ldots\right) + iA\left(x - \frac{1}{3!}x^3 + \ldots\right). \end{aligned}$$

Note that the series inside the brackets correspond to the Taylor expansions of cos and sin:

$$\begin{aligned} \cos(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots \\ \sin(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots \end{aligned}$$

In summary,

$$e^{iAx} = \cos(x)\mathbb{I} + i\sin(x)A.$$

□

Unitary matrix exponentials If $A \in \mathbb{C}^{n\times n}$ ($n \in \mathbb{N}$) is Hermitian, then e^{iA} is unitary.

A is called **Hermitian** if $A^\dagger = A$, with $A^\dagger = (A^*)^T$ denoting the conjugate transpose. $U \in \mathbb{C}^{n\times n}$ is called **unitary** if $U^\dagger U = \mathbb{I}$.

Proof. We first observe that

$$\left(e^{iA}\right)^\dagger = \sum_{k=0}^{\infty} \frac{1}{k!} \left((iA)^k\right)^\dagger = \sum_{k=0}^{\infty} \frac{1}{k!} \left(-iA^\dagger\right)^k = \sum_{k=0}^{\infty} \frac{1}{k!}(-iA)^k = e^{-iA}.$$

For any square matrices A and B which *commute* ($[A, B] = AB - BA = 0$), it holds that $e^A e^B = e^{A+B}$. (This can be shown analogous to the scalar case, but does not hold in general if $[A, B] \neq 0$.) Thus

$$\left(e^{iA}\right)^\dagger e^{iA} = e^{-iA} e^{iA} = e^{-iA+iA} = e^0 = \mathbb{I}.$$

Here we have used that $-iA$ commutes with iA (since these matrices only differ by a sign factor). □

Single-qubit rotation operators Pauli matrices X, Y, Z are Hermitian and self-inverse. Can be used to define the following unitary *rotation operators*, with parameter $\theta \in \mathbb{R}$:

$$R_x(\theta) := e^{-i\theta X/2} = \cos(\theta/2)\mathbb{I} - i\sin(\theta/2)X = \begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$$

$$R_y(\theta) := e^{-i\theta Y/2} = \cos(\theta/2)\mathbb{I} - i\sin(\theta/2)Y = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$$

$$R_z(\theta) := e^{-i\theta Z/2} = \cos(\theta/2)\mathbb{I} - i\sin(\theta/2)Z = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}$$

General case: rotation about an axis $\vec{v} \in \mathbb{R}^3$ (normalized such that with $\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2} = 1$):

Notation

$$\underbrace{\vec{v} \cdot \vec{\sigma}}_{\langle \vec{v} | \vec{\sigma} \rangle} \equiv v_1 X + v_2 Y + v_3 Z = \begin{pmatrix} v_3 & v_1 - iv_2 \\ v_1 + iv_2 & -v_3 \end{pmatrix} \in \mathbb{C}^{2\times 2}$$

It holds that $\vec{v} \cdot \vec{\sigma}$ is Hermitian and $(\vec{v} \cdot \vec{\sigma})^2 = \mathbb{I}$:

$$\begin{aligned}(\vec{v} \cdot \vec{\sigma})^2 &= \begin{pmatrix} v_3^2 + (v_1 - iv_2)(v_1 + iv_2) & v_3(v_1 - iv_2) + (v_1 - iv_2)(-v_3) \\ (v_1 + iv_2)v_3 - v_3(v_1 + iv_2) & v_3^2 + (v_1 - iv_2)(v_1 + iv_2) \end{pmatrix} \\ &= \begin{pmatrix} v_3^2 + v_1^2 + v_2^2 & 0 \\ 0 & v_3^2 + v_1^2 + v_2^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},\end{aligned}$$

where in the last step we have used that $\vec{v}$ is a unit vector, and therefore $\|\vec{v}\|^2 = v_1^2 + v_2^2 + v_3^2 = 1$.

General unitary *rotation operator about axis* $\vec{v}$ with angle $\theta \in \mathbb{R}$:

$$R_{\vec{v}}(\theta) := e^{-i\theta(\vec{v}\cdot\vec{\sigma})/2} = \cos(\theta/2)\mathbb{I} - i\sin(\theta/2)(\vec{v} \cdot \vec{\sigma}).$$

Note: R_x, R_y, R_z are special cases, corresponding to $\vec{v} = (1, 0, 0)$, $\vec{v} = (0, 1, 0)$, $\vec{v} = (0, 0, 1)$.

Can derive that Bloch sphere representation of $R_{\vec{v}}(\theta)$ is a "conventional" rotation (in three dimensions) by angle θ about axis $\vec{v}$ (see Tutorial 11): with $\vec{r}$ the Bloch vector of $|\psi\rangle$,

$$|\psi'\rangle = R_{\vec{v}}(\theta)|\psi\rangle$$

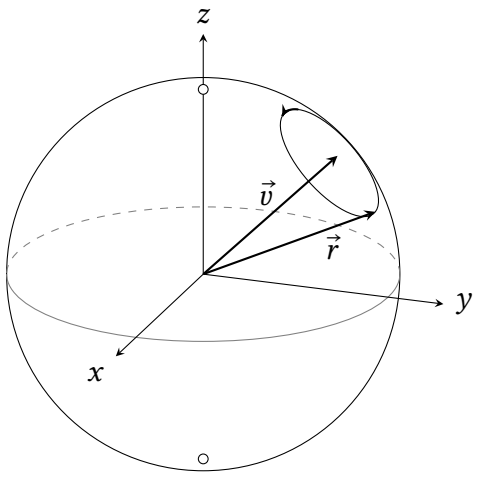


Figure 8: Rotation of the Bloch vector $\vec{r}$ about the axis $\vec{v}$.

3 Multiple qubits

So far we have only considered a single qubit, superpositions of basis states $|0\rangle$ and $|1\rangle$. For two qubits, this generalizes to

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$$

as computational basis states: all combinations (bit strings) of 0s and 1s. The generic two-qubit state is a superposition of these four basis states:

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle,$$

with amplitudes $\alpha_{ij} \in \mathbb{C}$, $i, j \in \{0, 1\}$, such that

$$|\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2 = 1$$

We can identify the basis states with vectors in $\mathbb{C}^4$:

$$|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad |10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad |11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Thus, a two-qubit state can be represented as:

$$|\psi\rangle = \begin{pmatrix} \alpha_{00} \\ \alpha_{01} \\ \alpha_{10} \\ \alpha_{11} \end{pmatrix} \in \mathbb{C}^4$$

See what happens if we measure only one qubit of a two-qubit state. For example, we measure the first qubit and we obtain the result

$$\begin{aligned} &0 \text{ with probability } |\alpha_{00}|^2 + |\alpha_{01}|^2 \\ &1 \text{ with probability } |\alpha_{10}|^2 + |\alpha_{11}|^2 \end{aligned}$$

Wavefunction directly after measurement

$$\begin{aligned} \text{if measurement is } 0 \ : \quad |\psi'\rangle &= \frac{\alpha_{00}|00\rangle + \alpha_{01}|01\rangle}{\sqrt{|\alpha_{00}|^2 + |\alpha_{01}|^2}} \\ \text{if measurement is } 1 \ : \quad |\psi'\rangle &= \frac{\alpha_{10}|10\rangle + \alpha_{11}|11\rangle}{\sqrt{|\alpha_{10}|^2 + |\alpha_{11}|^2}} \end{aligned}$$

where the denominator is the normalization factor. We now introduce the mathematical tool to construct two-qubit states: the tensor product of vector spaces.

3.1 Tensor products of vector spaces

Can combine two (arbitrary) vector spaces V and W to form the **tensor product** $V \otimes W$. The elements of $V \otimes W$ are linear combinations of "tensor products" $|v\rangle \otimes |w\rangle$ consisting of elements $|v\rangle \in V$ and $|w\rangle \in W$.

Example 3.1. Let $V = \mathbb{C}^2$, $W = \mathbb{C}^2$ be the single qubit spaces with basis $\{|0\rangle, |1\rangle\}$, then

$$\frac{1}{2} \underbrace{|0\rangle \otimes |0\rangle}_{=|00\rangle} + \frac{5i}{7} \underbrace{|1\rangle \otimes |0\rangle}_{=|10\rangle}$$

is an element of $V \otimes W$.

Let $\{|i\rangle_V : i = 1, \dots, m\}$ be a basis of V, and $\{|j\rangle_W : j = 1, \dots, n\}$ be a basis of W, then

$$\{|i\rangle_V \otimes |j\rangle_W : i = 1, \dots, m,\ j = 1, \dots, n\}$$

is a basis of $V \otimes W$. In particular:

$$\dim(V \otimes W) = \dim(V) \cdot \dim(W).$$

Note: $|i\rangle_V \otimes |j\rangle_W$ is also written as $|i\rangle|j\rangle$ or $|ij\rangle$.

Example 3.2. The basis construction for $V = \mathbb{C}^2$, $W = \mathbb{C}^2$ leads to the standard two-qubit basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$.

Elementary properties of tensor products:

- for all $|v\rangle \in V$, $|w\rangle \in W$ and $\alpha \in \mathbb{C}$:

$$\alpha(|v\rangle \otimes |w\rangle) = (\alpha|v\rangle) \otimes |w\rangle = |v\rangle \otimes (\alpha|w\rangle)$$

- for all $|v_1\rangle, |v_2\rangle \in V$ and $|w\rangle \in W$:

$$(|v_1\rangle + |v_2\rangle) \otimes |w\rangle = |v_1\rangle \otimes |w\rangle + |v_2\rangle \otimes |w\rangle$$

- for all $|v\rangle \in V$ and $|w_1\rangle, |w_2\rangle \in W$:

$$|v\rangle \otimes (|w_1\rangle + |w_2\rangle) = |v\rangle \otimes |w_1\rangle + |v\rangle \otimes |w_2\rangle$$

Vector notation (using standard basis), e.g.,

$$|v\rangle = v_1|0\rangle + v_2|1\rangle = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}, \quad |w\rangle = w_1|0\rangle + w_2|1\rangle = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix},$$

then:

$$\begin{aligned} |v\rangle \otimes |w\rangle &= (v_1|0\rangle + v_2|1\rangle) \otimes (w_1|0\rangle + w_2|1\rangle) \\ &= v_1 w_1|00\rangle + v_1 w_2|01\rangle + v_2 w_1|10\rangle + v_2 w_2|11\rangle \end{aligned}$$

Thus:

$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \otimes \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} v_1 w_1 \\ v_1 w_2 \\ v_2 w_1 \\ v_2 w_2 \end{pmatrix}$$

Example 3.3.

$$\begin{pmatrix}2\\3\end{pmatrix}\otimes\begin{pmatrix}5\\7\end{pmatrix}=\begin{pmatrix}10\\14\\15\\21\end{pmatrix}$$

Note: not every element of $V\otimes W$ can be written in the form $|v\rangle\otimes|w\rangle$ with $|v\rangle\in V$ and $|w\rangle\in W$, for example the so-called **Bell state**

$$|\psi\rangle=\frac{1}{\sqrt{2}}(|00\rangle+|11\rangle).$$

Assuming that V and W are equipped with an inner (scalar) product $\langle\cdot\,|\,\cdot\rangle$, one can define an inner product on $V\otimes W$ by:

$$\Big\langle\sum_j\alpha_j|v_j\rangle|w_j\rangle\;\Big|\;\sum_k\beta_k|\tilde{v}_k\rangle|\tilde{w}_k\rangle\Big\rangle:=\sum_j\sum_k\alpha_j^*\beta_k\langle v_j\,|\,\tilde{v}_k\rangle\langle w_j\,|\,\tilde{w}_k\rangle.$$

3.2 Generalization to n-qubits: 2^n computational basis states

$$\{\underbrace{|0,\dots,0\rangle}_{\text{length }n},|0,\dots,1\rangle,\dots,|1,\dots,1\rangle\}$$

Thus, a general n-qubit quantum state also denoted as "Quantum Register" is given by:

$$|\psi\rangle=\sum_{x_0=0}^{1}\sum_{x_1=0}^{1}\cdots\sum_{x_{n-1}=0}^{1}\alpha_{x_0,x_1,\dots,x_{n-1}}|x_0,x_1,\dots,x_{n-1}\rangle=\sum_{x=0}^{2^n-1}\alpha_x|x\rangle$$

where x is the integer with binary digits $x_0x_1\dots x_{n-1}$, and $\alpha_x\in\mathbb{C}$ for all $x\in\{0,1,\dots,2^n-1\}$ such that

$$\||\psi\rangle\|^2=\sum_{x=0}^{2^n-1}|\alpha_x|^2=1\quad\text{(normalization)}$$

In general, since the number 2^n of amplitudes grows exponentially with n, this is "hard" to simulate on a classical computer (for large n). This effect is known as the "curse of dimensionality".

We can see the vector space as a tensor product:

$$\underbrace{\mathbb{C}^2\otimes\mathbb{C}^2\otimes\dots\otimes\mathbb{C}^2}_{n\text{ times}}=(\mathbb{C}^2)^{\otimes n}=\mathbb{C}^{2^n}$$

3.3 Multi-qubit gates

As for single-qubit, an operation on multiple qubits is described by a unitary matrix $\mathcal{U}$. For n qubits, $\mathcal{U}\in\mathbb{C}^{2^n\times 2^n}$.

Example 3.4 (Controlled-NOT (CNOT) gate)**.** The CNOT gate is a two-qubit gate and it has a control qubit and a target qubit. The target qubit is flipped if the control qubit is in state $|1\rangle$, otherwise it is left unchanged.

$$|00\rangle\mapsto|00\rangle,\quad|01\rangle\mapsto|01\rangle,\quad|10\rangle\mapsto|11\rangle,\quad|11\rangle\mapsto|10\rangle$$

with the control qubit (first, in red) and the target qubit (second).

It can be expressed as:

$$|a, b\rangle \mapsto |a, a \oplus b\rangle \quad \text{for } a, b \in \{0, 1\}$$

where $\oplus$ denotes addition modulo 2.

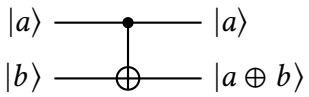


Figure 9: Circuit notation for the CNOT gate.

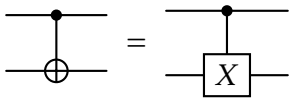
Matrix representation (in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$):

$$\mathcal{U}_{\text{CNOT}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Pauli-X

$\mathcal{U}_{\text{CNOT}}$ is unitary: it is real and symmetric, and $\mathcal{U}_{\text{CNOT}}^2 = \mathbb{I}$ (applying it twice flips the target back), so $\mathcal{U}_{\text{CNOT}}^\dagger \mathcal{U}_{\text{CNOT}} = \mathcal{U}_{\text{CNOT}}^2 = \mathbb{I}$.

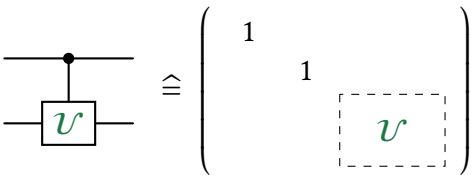
Alternative circuit notation:



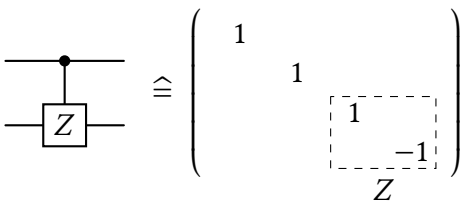
Controlled-$\mathcal{U}$ gates We can generalize Pauli-X to any unitary operator $\mathcal{U} \in \mathbb{C}^{2\times 2}$ acting on the target qubit, which gives the **controlled-$\mathcal{U}$ gate**:

$$|00\rangle \mapsto |00\rangle, \quad |01\rangle \mapsto |01\rangle, \quad |10\rangle \mapsto |1\rangle \otimes (\mathcal{U}|0\rangle), \quad |11\rangle \mapsto |1\rangle \otimes (\mathcal{U}|1\rangle).$$

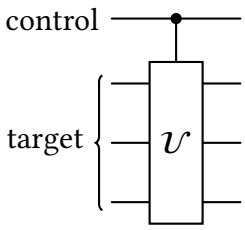
In circuit and matrix form:



Example 3.5 (Controlled-Z gate)**.** For $\mathcal{U} = Z$:



Controlled-$\mathcal{U}$ for multiple target qubits The control qubit can also act on several target qubits, with $\mathcal{U}$ a unitary on all of them:



Remark. Single-qubit and CNOT gates are **universal**: they can be used to implement an arbitrary unitary operation on n qubits (quantum analog of the universality of the classical NAND gate).

Example 3.6 (A circuit of single-qubit gates and CNOTs)**.** A circuit on three qubits consisting only of single-qubit gates and CNOTs:

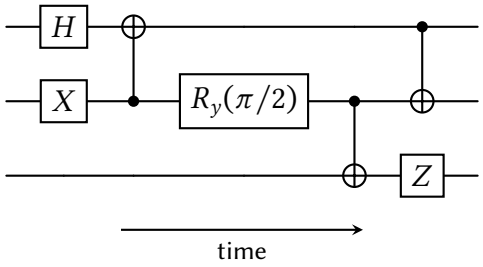
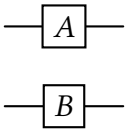


Figure 10: A circuit consisting only of single-qubit gates and CNOTs.

Kronecker products Matrix representation of single-qubit gates acting in parallel:



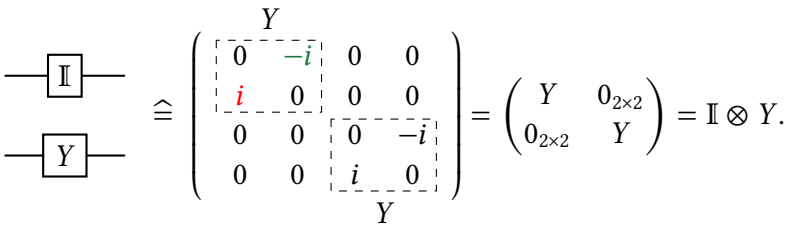
On the basis states $|a, b\rangle = |a\rangle \otimes |b\rangle$, $a, b \in \{0, 1\}$:

$$|a, b\rangle \mapsto (A|a\rangle) \otimes (B|b\rangle) = (A \otimes B)|a, b\rangle.$$

Example 3.7 ($A = \mathbb{I}$, $B = Y$)**.**

$$\begin{aligned}
|00\rangle &\mapsto |0\rangle \otimes (\underbrace{Y|0\rangle}_{i|1\rangle}) = i|01\rangle, \\
|01\rangle &\mapsto |0\rangle \otimes (\underbrace{Y|1\rangle}_{-i|0\rangle}) = -i|00\rangle, \\
|10\rangle &\mapsto |1\rangle \otimes (Y|0\rangle) = i|11\rangle, \\
|11\rangle &\mapsto |1\rangle \otimes (Y|1\rangle) = -i|10\rangle.
\end{aligned}$$

Matrix representation (the images of the basis states are the columns):



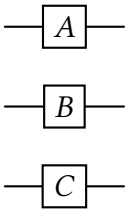
$$\widehat{=} \begin{pmatrix} 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \end{pmatrix} = \begin{pmatrix} Y & 0_{2\times 2} \\ 0_{2\times 2} & Y \end{pmatrix} = \mathbb{I} \otimes Y.$$

In general, the matrix representation of the tensor product of operators is the **Kronecker product**: for $A \in \mathbb{C}^{m\times n}$, $B \in \mathbb{C}^{p\times q}$,

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{pmatrix} \in \mathbb{C}^{mp\times nq}.$$

In NumPy: `np.kron(A, B)`.

This generalizes to an arbitrary number of tensor factors, e.g.



$$\widehat{=}\ A \otimes B \otimes C = (A \otimes B) \otimes C = A \otimes (B \otimes C),$$

where the ordering has to be kept, since $A \otimes B \neq B \otimes A$ in general.

Properties of the Kronecker product

(a) $(A \otimes B)^* = A^* \otimes B^*$ (elementwise complex conjugation)

(b) $(A \otimes B)^T = A^T \otimes B^T$ (transposition)

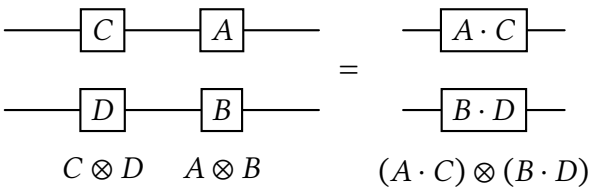
(c) $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$ (adjoint)

(d) $(A \otimes B) \otimes C = A \otimes (B \otimes C)$ (associative property)

(e) $(A \otimes B) \cdot (C \otimes D) = \underbrace{(A \cdot C) \otimes (B \cdot D)}_{\text{computationally faster than } (A\otimes B)\cdot(C\otimes D)}$ for matrices with compatible dimensions

($\cdot$: matrix-matrix multiplication)

In circuit notation, where $C \otimes D$ acts first:



(f) The Kronecker product of Hermitian matrices is Hermitian (by (c)).

(g) The Kronecker product of unitary matrices is unitary (follows from (c) and (e)):

$$(A \otimes B)^\dagger (A \otimes B) = (A^\dagger \otimes B^\dagger)(A \otimes B) = (A^\dagger A) \otimes (B^\dagger B) = \mathbb{I} \otimes \mathbb{I} = \mathbb{I}.$$

4 Quantum measurements

Recall the measurement of a single qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with respect to the computational basis $\{|0\rangle, |1\rangle\}$ (fig. 1): the measurement outputs classical data, the measurement outcome 0 or 1, on a double wire; in the alternative notation the single wire carries the qubit after the measurement (wavefunction collapse) and the measurement outcome leaves on the vertical double wire.

Unitary freedom of choice of measurement basis Given an orthonormal basis $\{|u_1\rangle, |u_2\rangle\}$, we can measure with respect to this basis by performing a base change before and after the measurement, with

$$\mathcal{U} = \big(|u_1\rangle, |u_2\rangle\big) \in \mathbb{C}^{2\times 2} \quad \text{unitary}$$

(the matrix with columns $|u_1\rangle$ and $|u_2\rangle$):

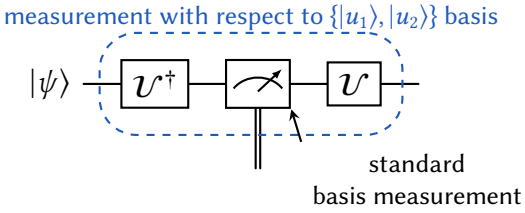
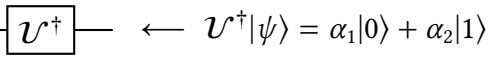


Figure 11: Change of measurement basis.

Representing a qubit as $|\psi\rangle = \alpha_1|u_1\rangle + \alpha_2|u_2\rangle$, $\alpha_1, \alpha_2 \in \mathbb{C}$, then



since $\mathcal{U}|0\rangle = |u_1\rangle$ and $\mathcal{U}|1\rangle = |u_2\rangle$ give $\mathcal{U}^\dagger|u_1\rangle = |0\rangle$ and $\mathcal{U}^\dagger|u_2\rangle = |1\rangle$. We will obtain the measurement result

$$\overset{\text{“}u_1\text{”}}{0} \quad \text{or} \quad \overset{\text{“}u_2\text{”}}{1} \quad \text{with respective probabilities } |\alpha_1|^2 \text{ and } |\alpha_2|^2.$$

After the measurement and applying $\mathcal{U}$, the final quantum state is

$$|u_1\rangle = \mathcal{U}|0\rangle \quad \text{or} \quad |u_2\rangle = \mathcal{U}|1\rangle.$$

4.1 General definition of quantum measurements

Definition 4.1. Quantum measurements are described by a collection $\{M_m\}$ of **measurement operators** acting on the quantum system, with the index m labelling possible measurement outcomes. Denoting the quantum state before the measurement by $|\psi\rangle$, result m occurs with probability

$$p(m) = \langle \psi \,|\, M_m^\dagger M_m \,|\, \psi \rangle = \| M_m|\psi\rangle \|^2 \qquad \big(\|v\|^2 = \langle v \,|\, v\rangle\big),$$

and the state after the measurement is

$$|\psi'\rangle = \frac{M_m|\psi\rangle}{\| M_m|\psi\rangle \|}.$$

The measurement operators satisfy the **completeness relation**

$$\sum_m M_m^\dagger M_m = \mathbb{I}.$$

The completeness relation ensures that the probabilities sum to 1:

$$\sum_m p(m) = \sum_m \langle\psi \mid M_m^\dagger M_m \mid \psi\rangle = \Big\langle \psi \,\Big|\, \underbrace{\Big(\sum_m M_m^\dagger M_m\Big)}_{\mathbb{I}} \,\Big|\, \psi \Big\rangle = \langle\psi \mid \psi\rangle = 1.$$

Example 4.2 (Measurement in the computational basis)**.** Measurement of a qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with respect to the computational basis $\{|0\rangle, |1\rangle\}$:

$$M_0 := \underbrace{|0\rangle\langle 0|}_{\text{outer product}} = \underbrace{\begin{pmatrix}1\\0\end{pmatrix}}_{2\times 1} \cdot \underbrace{\begin{pmatrix}1 & 0\end{pmatrix}}_{1\times 2} = \begin{pmatrix}1 & 0\\0 & 0\end{pmatrix},$$

$$M_1 := |1\rangle\langle 1| = \begin{pmatrix}0\\1\end{pmatrix}\begin{pmatrix}0 & 1\end{pmatrix} = \begin{pmatrix}0 & 0\\0 & 1\end{pmatrix}.$$

Then

$$p(0) = \langle\psi \mid M_0^\dagger M_0 \mid \psi\rangle = \langle\psi \mid M_0 \mid \psi\rangle = \begin{pmatrix}\alpha^* & \beta^*\end{pmatrix}\begin{pmatrix}1 & 0\\0 & 0\end{pmatrix}\begin{pmatrix}\alpha\\\beta\end{pmatrix} = |\alpha|^2,$$

$$p(1) = \langle\psi \mid M_1^\dagger M_1 \mid \psi\rangle = \cdots = |\beta|^2,$$

where $M_0^\dagger M_0 = M_0$ since M_0 is Hermitian and $M_0^2 = M_0$.

4.2 Orthogonal projection operators

Definition 4.3. A square matrix $P \in \mathbb{C}^{n\times n}$ is called an **orthogonal projection matrix** if it is Hermitian ($P^\dagger = P$) and satisfies $P^2 = P$.

Let V be a k-dimensional subspace of $\mathbb{C}^n$, and $\{|u_1\rangle, \ldots, |u_k\rangle\}$ an orthonormal ($\langle u_i \mid u_j\rangle = \delta_{ij}$) basis of V (which always exists). Then

$$P = \sum_{j=1}^{k} |u_j\rangle\langle u_j|$$

(with $\langle u_j| = |u_j\rangle^\dagger$) is the **projector** onto V, acting on a vector $|w\rangle \in \mathbb{C}^n$ as

$$P|w\rangle = \sum_{j=1}^{k} |u_j\rangle\langle u_j \mid w\rangle.$$

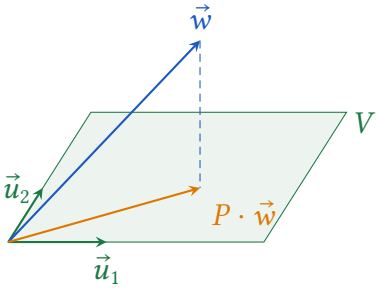


Figure 12: Orthogonal projection onto a subspace.

It is straightforward to show that P is an orthogonal projection matrix, that P is independent of the chosen basis, and that $P|v\rangle = |v\rangle$ for all $|v\rangle \in V$.

$$P^2 = \Big(\sum_{j=1}^{k} |u_j\rangle\langle u_j|\Big)\Big(\sum_{j'=1}^{k} |u_{j'}\rangle\langle u_{j'}|\Big) = \sum_{j=1}^{k}\sum_{j'=1}^{k} |u_j\rangle \underbrace{\langle u_j \,|\, u_{j'}\rangle}_{\delta_{jj'}} \langle u_{j'}| = \sum_{j=1}^{k} |u_j\rangle\langle u_j| = P.$$

Relation to spectral decomposition Every **normal** matrix $A \in \mathbb{C}^{n\times n}$ (i.e., $AA^\dagger = A^\dagger A$) is diagonalizable by an orthonormal basis, that is, there exist a unitary $U \in \mathbb{C}^{n\times n}$ and eigenvalues $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$ such that

$$A = U \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} U^\dagger = \sum_{j=1}^{n} \lambda_j |u_j\rangle\langle u_j|$$

where the $|u_j\rangle$ are the column vectors of U: $U = \big(u_1\big|u_2\big|\cdots\big|u_n\big)$.

Some eigenvalues can coincide, for example $\lambda_1 = \lambda_2$; then

$$\lambda_1 |u_1\rangle\langle u_1| + \lambda_2 |u_2\rangle\langle u_2| = \lambda_1 P,$$

with $P = |u_1\rangle\langle u_1| + |u_2\rangle\langle u_2|$ the projector onto the (two-dimensional) eigenspace corresponding to eigenvalue λ_1. In general, we can write the spectral decomposition as

$$A = \sum_m \lambda_m P_m$$

where the sum is over all *distinct* eigenvalues and the P_m are the projectors onto the corresponding eigenspaces.

4.3 Projective measurements

Definition 4.4. A **projective measurement** is described by an **observable** M, a Hermitian operator acting on the quantum system, with spectral decomposition (M is Hermitian, hence normal)

$$M = \sum_m \lambda_m P_m,$$

with P_m the projection onto the eigenspace with eigenvalue λ_m. The possible outcomes of the measurement correspond to the eigenvalues λ_m.

Probability of obtaining result λ_m:

$$p(\lambda_m) = \langle\psi \,|\, \underbrace{P_m}_{\substack{= P_m^\dagger P_m = P_m^2 = P_m \\ \text{since } P_m \text{ is a projection}}} \,|\, \psi\rangle = \langle\psi \,|\, P_m^\dagger P_m \,|\, \psi\rangle = \|P_m|\psi\rangle\|^2.$$

State of the quantum system after the measurement:

$$|\psi'\rangle = \frac{P_m|\psi\rangle}{\|P_m|\psi\rangle\|} = \frac{P_m|\psi\rangle}{\sqrt{p(\lambda_m)}}.$$